May 2026

amazon

AMAZON
SALES ANALYSIS

BUSINESS INTELLIGENCE REPORT

An in-depth SQL & Power BI driven analysis to discover key insights, understand customer behavior and drive data-informed business decisions.

TOOLS & TECHNOLOGIES USED

- MySQL
- Excel
- Power BI
- Data Analysis

KEY BUSINESS HIGHLIGHTS

128.97K Total Orders Overall number of orders	**₹78.59M** Total Revenue Total revenue generated	**₹609.40** Avg. Order Value Average amount spent per order	**₹999** Highest Order Value Highest single order value recorded
69.55% Amazon Fulfilment Orders fulfilled by Amazon	**14.21%** Cancelled Orders Orders that were cancelled	**99.32%** B2C Orders Business-to-Consumer order share	Top State **Maharashtra** Highest revenue contribution

WHAT THIS REPORT COVERS

- Order & Sales Performance Analysis
- Category, State & Channel Insights
- Fulfilment & Cancellation Analysis
- Promotion Effectiveness Analysis
- Order Value & Quantity Analysis
- Data-Driven Business Recommendations

BUSINESS IMPACT

- Identify high-performing categories
- Optimize fulfilment & reduce cancellations
- Understand customer purchasing behavior
- Improve revenue through targeted promotions
- Support strategic decision making with data-driven insights

PREPARED BY

ARYAN
Business Analyst

“Turning data into insights and insights into impact.”

Data-Driven Insights | Strategic Decisions | Business Growth | Operational Efficiency

Data | Insights | Impact

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📊 1. Executive Summary

This project presents a comprehensive analysis of Amazon sales data using **Excel, MySQL, and Power BI** to uncover actionable business insights. The objective was to evaluate sales performance, customer purchasing behaviour, fulfilment efficiency, promotional effectiveness, and regional trends to support data-driven decision-making.

The dataset consists of **128,975 customer orders** with a total revenue of **₹78.59 million** and an average order value of **₹609.36**. The analysis identified key performance indicators, including fulfilment trends, cancellation rates, category performance, state-wise sales distribution, and customer purchase patterns.

SQL was used to clean, transform, and analyse the dataset, while Power BI was used to develop an interactive dashboard for visualizing business performance. The findings highlight opportunities to improve fulfilment efficiency, optimize promotional campaigns, reduce order cancellations, and focus on high-performing products and regions.

Overall, this project demonstrates how data analytics can transform raw transactional data into meaningful business insights, enabling organizations to make informed strategic and operational decisions.

📌 *Key Highlights*

- *128,975 total customer orders analysed*
- *₹78.59 Million total revenue generated*
- *₹609.36 average order value*
- *69.55% Amazon Fulfilment*
- *99.32% B2C orders*
- *14.21% order cancellation rate*

📌 2. Project Overview

Objective

The primary objective of this project is to analyze Amazon sales data and extract meaningful business insights using modern data analytics tools. The analysis focuses on identifying sales trends, customer purchasing behaviour, fulfilment performance, promotional effectiveness, and regional sales patterns to support informed business decisions.

Project Scope

This project covers the complete data analytics workflow, beginning with raw data preparation and ending with interactive business intelligence dashboards. The analysis includes:

- Sales and revenue performance analysis
- Product category performance
- State-wise sales analysis
- Sales channel analysis
- Order status and fulfilment analysis
- Promotion effectiveness analysis
- Customer purchasing behaviour
- Business recommendations based on analytical findings

Business Problem

E-commerce businesses generate large volumes of transactional data every day. Without proper analysis, valuable insights remain hidden, making it difficult to identify high-performing products, customer trends, fulfilment bottlenecks, and revenue opportunities. This project demonstrates how data analytics can convert raw sales data into actionable business intelligence.

📊 3. Dataset Overview

Dataset Information

Attribute	Details
Dataset	Amazon Sales Dataset
Total Orders	**128,975**
Total Revenue	**₹78,592,678.30**
Average Order Value	**₹609.36**
Total Quantity Sold	**116,649**
Number of Columns	**20**
File Format	CSV
Industry	E-commerce

Key Features Analysed:

- *Order Details*
- *Product Categories*
- *Sales Amount*
- *Quantity Ordered*
- *Fulfilment Method*
- *Order Status*
- *Sales Channel*
- *Customer Type (B2B/B2C)*
- *Promotions*
- *Geographic Information*

Data Quality

Before analysis, the dataset was reviewed for:

- Missing values
- Duplicate records
- Invalid data types
- Inconsistent formatting
- Null values
- Data standardization

The cleaned dataset was then imported into MySQL for SQL-based analysis and Power BI for visualization.

4. Methodology

The project follows a structured data analytics workflow consisting of four major stages.

1. Data Collection

- Imported Amazon sales dataset in CSV format.
- Verified dataset structure and attributes before analysis.

2. Data Cleaning (Excel)

The dataset was cleaned using Microsoft Excel to improve data quality.
Activities included:

- Removing duplicates
- Handling missing values
- Standardizing text formatting
- Validating numerical fields
- Preparing the dataset for SQL import

3. SQL Analysis (MySQL)

MySQL was used to perform business analysis through SQL queries.
The analysis covered:

- Sales Performance
- Category Analysis
- State Analysis
- Sales Channels
- Fulfilment Analysis
- Order Status Analysis
- Promotion Analysis
- Customer Behaviour
- Business Insights

4. Dashboard Development (Power BI)

Power BI was used to build an interactive dashboard to visualize key business metrics.
Dashboard components include:

- KPI Cards
- Sales Trends
- Category Performance
- State-wise Revenue
- Fulfilment Distribution
- Sales Channels
- Interactive Filters (Slicers)

5. Key Performance Indicators (KPIs)

Project Overview

The following KPIs summarize the overall performance of the Amazon sales dataset and provide a high-level view of key business metrics identified during the analysis.

KPI	Value
Total Orders	**128,975**
Total Revenue	**₹78.59 Million**
Average Order Value	**₹609.36**
Total Quantity Sold	**116,649**
Average Quantity per Order	**0.90**
Amazon Fulfilment	**69.55%**
Merchant Fulfilment	**30.45%**
B2C Orders	**99.32%**
B2B Orders	**0.68%**
Cancelled Orders	**14.21%**

Key Observations

- Amazon fulfilled **69.55%** of all customer orders, highlighting its dominant fulfilment network.
- The platform generated **₹78.59 million** in revenue from **128,975** orders.
- The average customer order value was **₹609.36**.
- Nearly all transactions (**99.32%**) were Business-to-Consumer (B2C) orders.
- The overall cancellation rate remained **14.21%**, indicating opportunities to improve fulfilment efficiency and customer satisfaction.

6. Business Analysis & Findings

Business Analysis & Findings

This section presents the findings obtained through SQL-based analysis of the Amazon sales dataset. Each subsection focuses on a specific business area, including sales performance, product categories, regional performance, fulfilment, promotions, customer purchasing behaviour, and order trends. The objective is to identify meaningful patterns, evaluate business performance, and provide actionable recommendations supported by data.

6.1 Sales Performance Analysis

Objective

To evaluate the overall sales performance of the business by analysing total orders, revenue generation, average order value, and sales volume. This analysis provides a high-level understanding of business growth and operational performance.

Key Performance Metrics

Metric	Value
Total Orders	**128,975**
Total Revenue	**₹78.59 Million**
Average Order Value	**₹609.36**
Total Quantity Sold	**116,649**
Average Quantity per Order	**0.90**

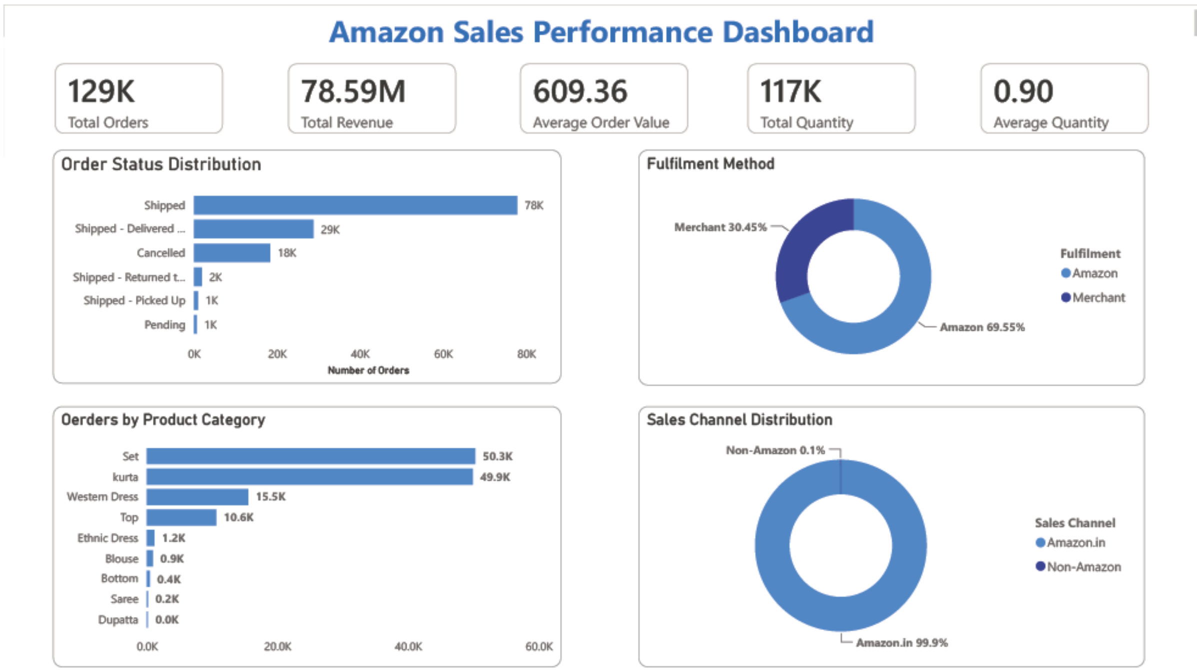
Key Findings

- The business processed **128,975 customer orders**, demonstrating a high transaction volume.
- Total revenue reached **₹78.59 million**, reflecting strong sales performance.
- Customers spent an average of **₹609.36** per order.
- A total of **116,649 units** were sold, with an average order quantity of **0.90 items**.

Business Interpretation

The sales metrics indicate consistent customer demand and a healthy revenue stream. The average order value suggests moderate customer spending, while the relatively low average quantity per order indicates that customers typically purchase a single item per transaction. These insights provide a baseline for evaluating future growth opportunities and improving customer purchase value.

Power BI Visualization



6.2 Product Category Analysis

Objective

To identify the best-performing product categories based on sales performance and order volume, enabling a better understanding of customer preferences and high-revenue product segments.

Category Performance

Category	Total Orders
Set	50,284
Kurta	49,877
Western Dress	15,500
Top	10,622
Ethnic Dress	1,159

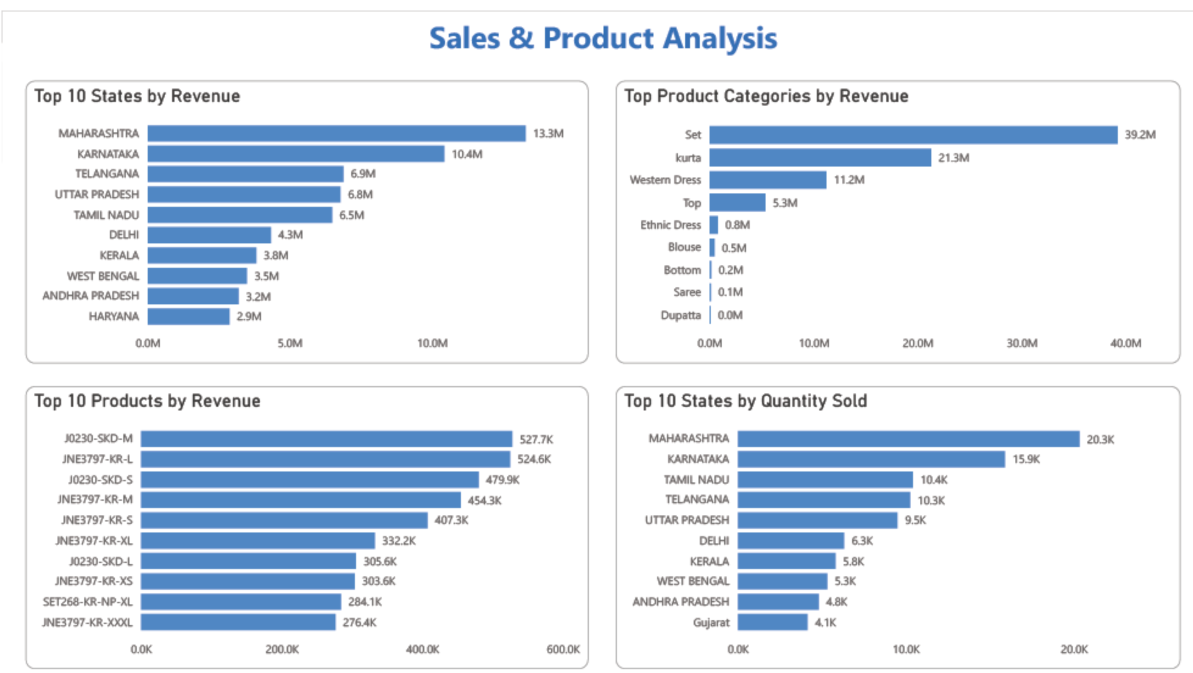
Key Findings

- **Set** was the highest-selling product category with **50,284 orders**.
- **Kurta** followed closely with **49,877 orders**, indicating consistently high customer demand.
- **Western Dress** ranked third with **15,500 orders**, significantly lower than the top two categories.
- The remaining categories contributed only a small proportion of total orders, showing a highly concentrated product mix.
- **Dupatta** recorded the lowest demand with only **3 orders**.

Business Interpretation

The analysis reveals that customer demand is heavily concentrated in the Set and Kurta categories, which together account for the majority of orders. This indicates strong customer preference for these products and suggests that inventory planning, promotional campaigns, and product expansion should prioritize these high-performing categories. Categories with consistently low order volumes may require targeted marketing efforts or a review of their product strategy.

Power BI Dashboard



🌍 6.3 State-wise Sales Analysis

Objective

To analyse revenue distribution across different states and identify the highest-performing regions, enabling better geographical targeting and business expansion strategies.

Key Findings

- **Maharashtra** generated the highest revenue (**₹13.34 million**), making it the strongest performing market.
- **Karnataka** ranked second with **₹10.48 million**, indicating strong customer demand.
- **Telangana**, **Uttar Pradesh**, and **Tamil Nadu** also contributed significantly, each generating more than **₹6.5 million** in revenue.
- The top five states accounted for a substantial share of overall sales, highlighting concentrated regional demand.
- Revenue declined gradually after the top five states, indicating opportunities to strengthen market presence in lower-performing regions.

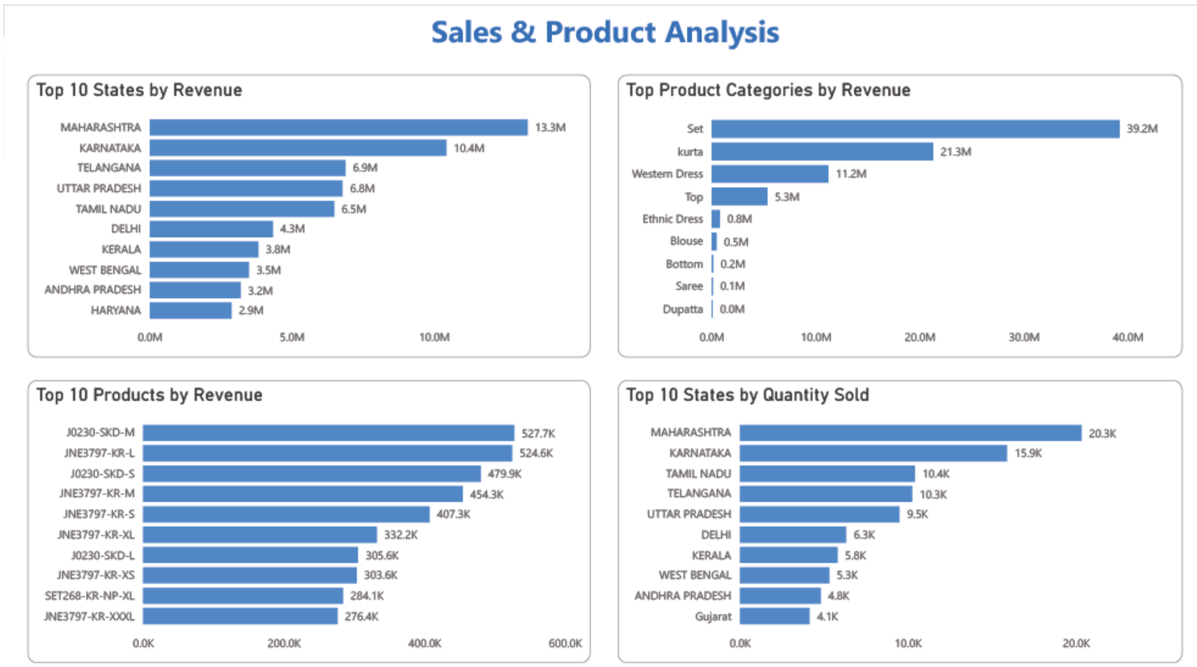
Top 10 States by Revenue

State	Revenue
Maharashtra	**₹13.34 M**
Karnataka	**₹10.48 M**
Telangana	**₹6.92 M**
Uttar Pradesh	**₹6.82 M**
Tamil Nadu	**₹6.52 M**
Delhi	**₹4.35 M**
Kerala	**₹3.83 M**
West Bengal	**₹3.51 M**
Andhra Pradesh	**₹3.22 M**
Haryana	**₹2.88 M**

Business Interpretation

The revenue distribution demonstrates that sales are concentrated in a few key states, with Maharashtra and Karnataka leading overall performance. These regions represent important markets for customer acquisition, inventory planning, and promotional campaigns. Lower-performing states present opportunities for targeted marketing initiatives and regional expansion strategies to improve market penetration and revenue growth.

Power BI Dashboard



🛒 6.4 Sales Channel Analysis

Objective

To analyse the distribution of customer orders across different sales channels and identify the platform contributing the highest order volume.

Sales Channel Performance

Sales Channel	Total Orders
Amazon.in	**128,851**
Non-Amazon	**124**

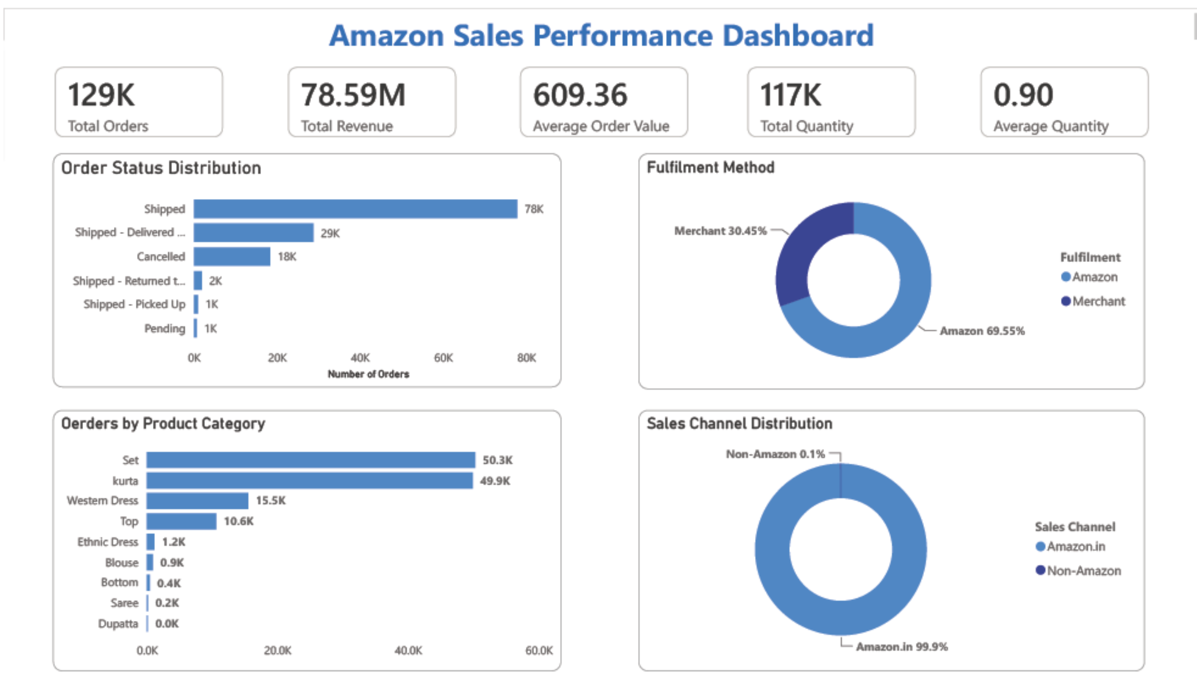
Key Findings

- **Amazon.in** accounted for **128,851 orders**, representing virtually all customer transactions.
- The **Non-Amazon** channel generated only **124 orders**, contributing a negligible share of total sales.
- Customer purchases were overwhelmingly concentrated on the Amazon marketplace.
- The results indicate a strong dependence on Amazon.in as the primary sales platform.

Business Interpretation

The analysis shows that the business relies almost entirely on **Amazon.in** for order generation, with the Non-Amazon channel contributing an insignificant number of sales. This highlights the effectiveness of Amazon's marketplace in reaching customers while also indicating potential opportunities to diversify sales channels and reduce dependence on a single platform.

Power BI Dashboard



🚚 6.5 Fulfilment Analysis

Objective

To evaluate the distribution of customer orders based on fulfilment methods and understand the operational dependence on Amazon and merchant fulfilment services.

Fulfilment Performance

Fulfilment Method	Total Orders
Amazon	**89,698**
Merchant	**39,277**

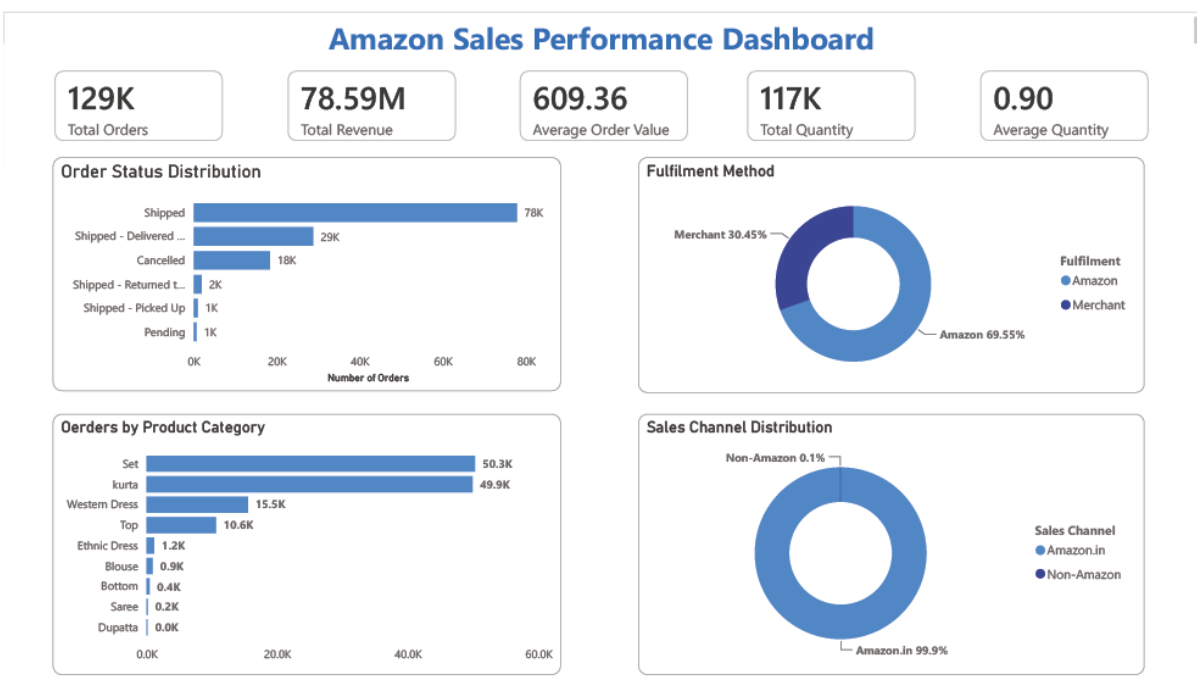
Key Findings

- **Amazon Fulfilment** processed **89,698 orders**, accounting for **69.55%** of all customer orders.
- **Merchant Fulfilment** handled **39,277 orders**, representing **30.45%** of total orders.
- More than two-thirds of all orders were fulfilled through Amazon's logistics network.
- Amazon Fulfilment emerged as the dominant fulfilment method, demonstrating a strong reliance on Amazon's delivery infrastructure.

Business Interpretation

The analysis indicates that Amazon's fulfilment network plays a significant role in order processing and customer delivery. The higher adoption of Amazon Fulfilment suggests greater operational efficiency and customer trust in Amazon's logistics capabilities. While Merchant Fulfilment contributes a considerable share of orders, further optimization of merchant operations could help improve delivery consistency and overall customer experience.

Power BI Dashboard



6.6 Order Status Analysis

Objective

To analyse the distribution of order statuses and evaluate the efficiency of the order fulfilment process by identifying completed, cancelled, returned, and pending orders.

Order Status Summary

Order Status	Total Orders
Shipped	**77,804**
Shipped – Delivered to Buyer	**28,769**
Cancelled	**18,332**
Shipped – Returned to Seller	**1,953**
Shipped – Picked Up	**973**
Pending	**658**
Pending – Waiting for Pick Up	**281**
Others*	**205**

*Others include Returning to Seller, Out for Delivery, Rejected by Buyer, Shipping, Lost in Transit, and Damaged.

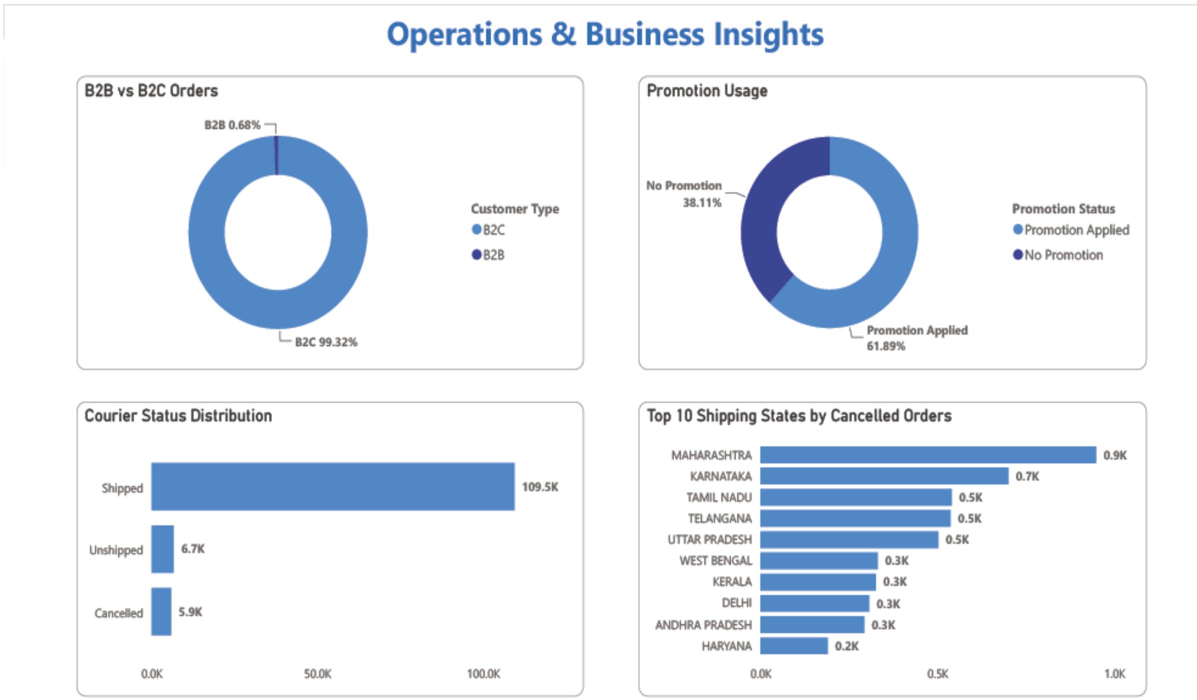
Key Findings

- **Shipped** was the most common order status, with **77,804 orders**, indicating a high volume of successfully processed shipments.
- **28,769 orders** were successfully delivered to customers.
- **18,332 orders** were cancelled, representing a **14.21% cancellation rate**.
- Returns and pending orders accounted for a relatively small proportion of total orders.
- Only a negligible number of orders were affected by issues such as damage, rejection, or loss in transit.

Business Interpretation

The order status distribution indicates that the fulfilment process is largely effective, with the majority of orders progressing through shipment and delivery stages. However, the cancellation volume highlights an opportunity to improve order confirmation, inventory availability, and fulfilment coordination. The low number of damaged and lost shipments suggests that the logistics network maintains a high level of operational reliability.

Power BI Dashboard



🎯 6.7 Promotion Analysis

Objective

To evaluate the impact of promotional campaigns by analysing the number of orders placed with and without promotional offers.

Promotion Performance

Promotion Status	**Total Orders**
Promotion Applied	**79,822**
No Promotion	**49,153**

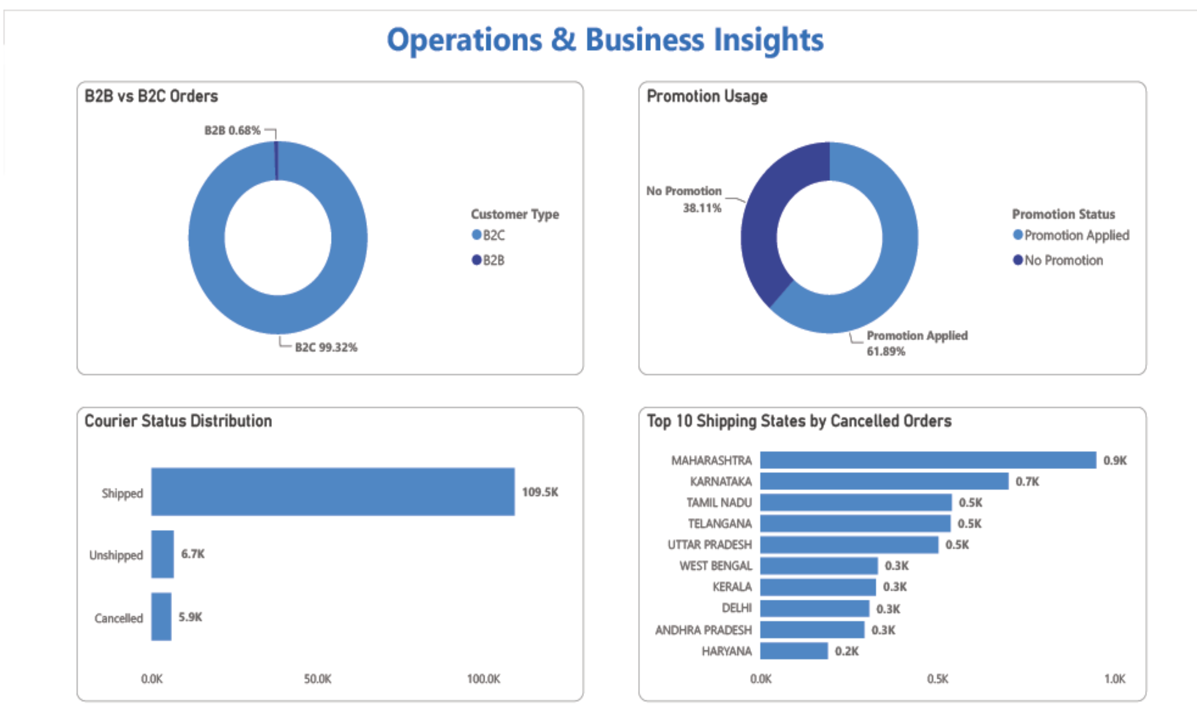
Key Findings

- **79,822 orders** were placed using promotional offers, representing the majority of total orders.
- **49,153 orders** were completed without any promotion.
- Promotional campaigns influenced a significant proportion of customer purchases.
- The results indicate that discounts and promotional offers play an important role in driving customer engagement and sales volume.

Business Interpretation

The analysis suggests that promotional campaigns are a major contributor to customer purchases. A substantial number of customers utilized promotional offers, highlighting the effectiveness of discounts in influencing buying behaviour. While promotions can increase order volume, businesses should continuously evaluate their profitability to ensure that promotional strategies generate sustainable revenue growth.

Power BI Dashboard



🛍 6.8 Customer Purchase Behaviour

Objective

To analyse customer purchasing patterns based on order quantity, order value, and purchasing preferences, helping identify customer buying behaviour and revenue contribution.

Customer Purchase Summary

Order Quantity Classification

Order Type	Total Orders
Single Item	**115,780**
Multiple Items	**13,195**

Order Value Classification

Order Value	Total Orders
Low Value Orders	**107,847**
High Value Orders	**13,333**

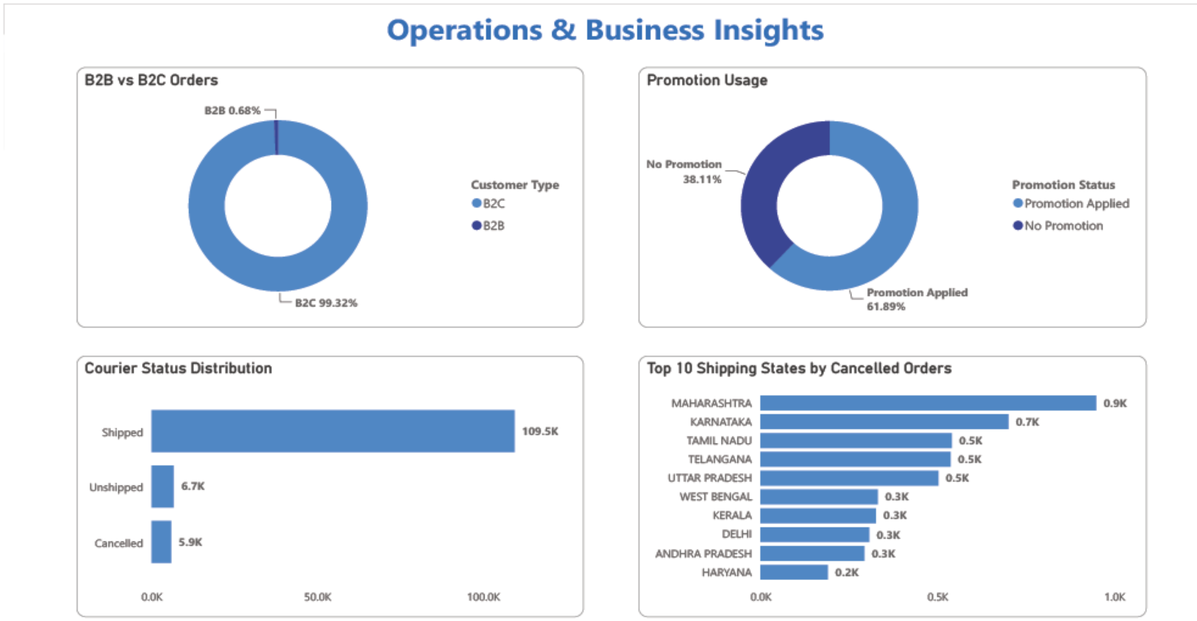
Key Findings

- **89.77%** of all orders were **single-item purchases**, while only **10.23%** contained multiple items.
- **Low-value orders** dominated the dataset with **107,847 orders**, representing the majority of customer transactions.
- **13,333 high-value orders** (₹1,000 and above) contributed significantly to overall revenue despite their lower frequency.
- The **Set** category recorded **12,221 high-value orders**, making it the leading contributor to premium purchases.
- Orders containing a single product generated the majority of overall sales activity, indicating a preference for individual product purchases.

Business Interpretation

Customer purchasing behaviour indicates a strong preference for single-item and low-value transactions. However, high-value purchases—particularly within the **Set** category—play a crucial role in revenue generation. This presents an opportunity to increase average order value through product bundling, cross-selling, and targeted promotional strategies encouraging customers to purchase multiple items in a single transaction.

Power BI Dashboard



📊 6.9 Advanced Business Insights

Objective

To derive advanced business insights using SQL CASE statements by analysing customer purchase patterns, revenue distribution, order value segments, and high-value product categories. These insights support strategic decision-making and identify opportunities for revenue growth.

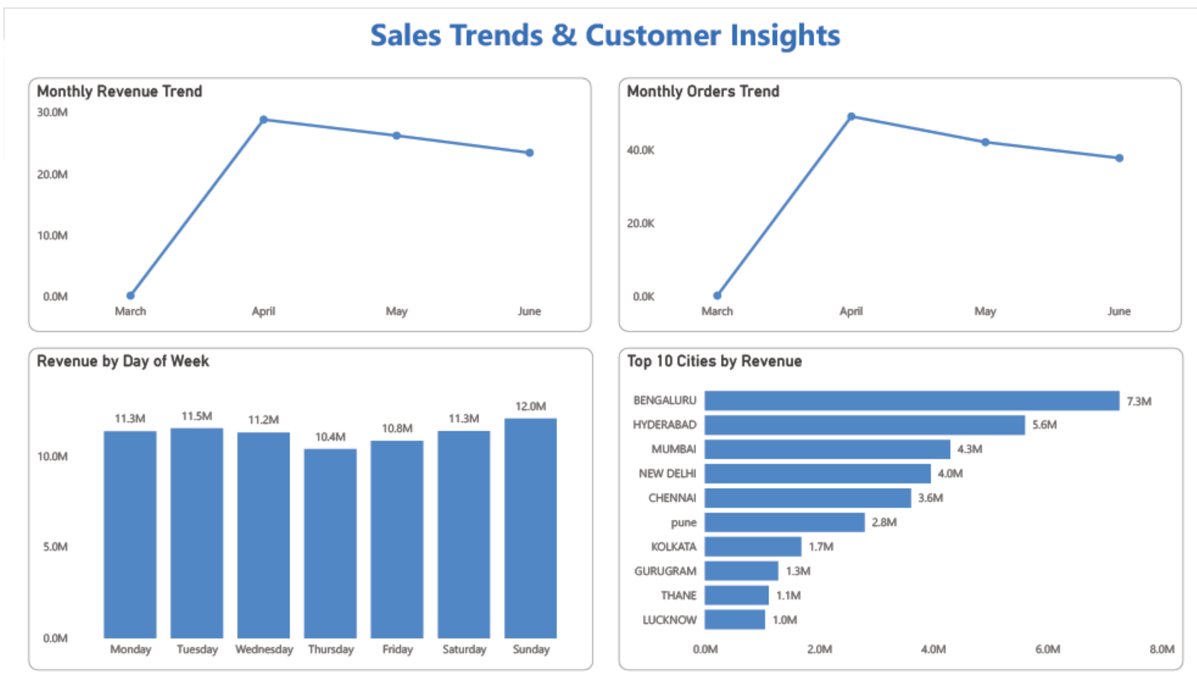
Key Findings

- **Low-value orders (< ₹1,000)** accounted for **107,847 orders**, while **13,333 orders** were classified as high-value purchases.
- Although fewer in number, **high-value orders generated ₹16.23 million** in revenue, making them an important contributor to overall sales.
- The **₹500–₹1,000** revenue bucket generated the **highest revenue (₹45.57 million)** and contained **64,214 orders**, indicating this as the strongest pricing segment.
- **Single-item purchases** dominated customer behaviour with **115,780 orders**, compared to **13,195 multiple-item purchases**.
- The **Set** category recorded **12,221 high-value orders**, significantly outperforming all other product categories in premium sales.

Business Interpretation

The advanced analysis reveals that while most customers place low-value, single-item orders, premium purchases contribute substantially to overall revenue. The ₹500–₹1,000 price range emerged as the most valuable segment, suggesting that customers are highly responsive to products within this pricing band. Furthermore, the dominance of the **Set** category among high-value purchases highlights a strong opportunity to expand premium offerings, introduce product bundles, and implement targeted upselling strategies to increase average order value.

Power BI Dashboard



7. Business Recommendations

Business Recommendations

Based on the analysis performed on the Amazon sales dataset, the following recommendations are proposed to improve business performance and customer experience.

1. Expand High-Performing Product Categories

The **Set** and **Kurta** categories generated the highest order volumes and contributed significantly to premium sales. Increasing inventory, expanding product variety, and introducing new designs within these categories can help drive additional revenue.

2. Strengthen Presence in High-Revenue States

Maharashtra, Karnataka, Telangana, Uttar Pradesh, and Tamil Nadu generated the highest revenue. Region-specific marketing campaigns and improved inventory planning in these states can further increase sales.

3. Optimize Promotional Campaigns

More than **79,000 orders** were placed using promotional offers, indicating that discounts significantly influence purchasing behaviour. Future campaigns should focus on improving profitability by targeting high-performing products and customer segments.

4. Increase Average Order Value

Since most customers purchase only a single item per order, introducing product bundles, cross-selling, and personalized recommendations can encourage customers to purchase multiple products in a single transaction.

5. Reduce Order Cancellations

The cancellation rate indicates an opportunity to improve inventory availability, order confirmation processes, and fulfilment efficiency. Addressing these issues can improve customer satisfaction and reduce revenue loss.

6. Promote Premium Product Segments

High-value orders contributed significantly to total revenue despite representing a smaller proportion of transactions. Expanding premium product offerings and targeted upselling strategies can further improve overall revenue.

📝 8. Conclusion

This project demonstrated the application of **Excel, MySQL, and Power BI** to analyse Amazon sales data and generate actionable business insights. Through data cleaning, SQL analysis, and interactive dashboard development, the project successfully identified key trends related to sales performance, product categories, customer purchasing behaviour, fulfilment methods, promotional effectiveness, and regional performance.

The analysis highlighted opportunities to improve fulfilment efficiency, optimize promotional strategies, increase average order value, and strengthen performance in high-revenue markets. These insights demonstrate how data analytics can support informed business decisions and improve operational performance.

Overall, the project showcases practical data analytics skills, including data preparation, SQL querying, business analysis, data visualization, and insight-driven decision-making.

📎 9. Appendix

Dataset Information

- Dataset: Amazon Sales Dataset
- Total Records: **128,975**
- File Format: CSV
- Industry: E-commerce

Tools & Technologies

Tool	Purpose
Microsoft Excel	Data Cleaning & Preparation
MySQL	Data Analysis & SQL Queries
Power BI	Dashboard & Data Visualization
Microsoft Word	Business Report Documentation

Key KPIs

- Total Orders
- Total Revenue
- Average Order Value
- Average Quantity per Order
- B2B vs B2C Distribution
- Amazon vs Merchant Fulfilment
- Promotion Analysis
- Order Status Analysis
- Customer Purchase Behaviour
- Regional Sales Performance

Analytical Techniques Used

- Data Cleaning
- Data Aggregation
- GROUP BY
- CASE Statements
- Aggregate Functions
- Business KPI Analysis
- Revenue Segmentation
- Customer Behaviour Analysis
- Data Visualization